

Unit-1 → water

② sources of water

A Surface water → Rain water
River water
Lake water
Sea water

B Underground water → (spring and well water).

③ Common natural impurities of water.

A Suspended impurities → Inorganic
Organic

B Colloidal impurities → Inorganic
Organic

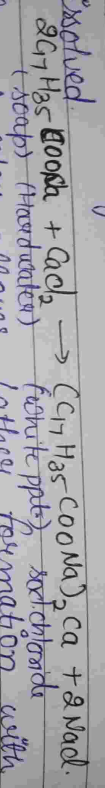
C Dissolved impurities → Salts
Gases

D Biological impurities

① Hardness of water:

Definition:-> Hardness is that property of the water which prevents lather formation with a soap solution.

Hard water is the water in which some kind of salt (Ca^{2+} , Mg^{2+} , Fe^{2+} , SO_4^{2-} and HCO_3^-) has been dissolved.



Note:-> (i) Normal water allows lather formation with soap.
(ii) Hard water prevents lather formation with soap.

→ There are two types of Hardness.

Temporary Hardness	Permanent Hardness
It is due to the presence of carbonates & bicarbonates (HCO_3^-) in water & can be easily eliminated by boiling.	Occurs due to the presence of the ions Ca^{2+} , Mg^{2+} , Fe^{3+} and SO_4^{2-} & cannot be eliminated by boiling & by chemical methods.
It is also known as carbonate hardness.	

Note:- Hardness is caused by the 'DIVALENT METALLIC CATION'.

① Degree of Hardness / Measurement of Hardness:

The degree of hardness of water is conveniently expressed in terms of equivalent amount of CaCO_3 .

→ choice of CaCO_3 as the standard for reporting hardness of water is due to:

1. Its molecular weight which is exactly 100 and equivalent weight is 50 (easy easy for calculation)
2. It is the most insoluble salt, that can be precipitated in water treatment processes.

② Formulae ⇒

Weight of the hardness
producing substance
(in mg/l)

x

Chemical equivalent
of CaCO_3 (i.e 50)

x 2

[Chemical equivalent of hardness producing
substance] x 2

$$= \frac{\text{Weight of the hardness producing substance (in mg/l)} \times 100}{\text{Chemical equivalent of hardness producing substance} \times 2}$$

(or)

→ Multiplication Factor (in mg/L or ppm) $\times$ weight of hardness producing substance (in mg/L)

$$\text{Multiplication Factor (M.F.)} = \frac{\text{Mol. weight of CaCO}_3}{2 \times \text{chemical equivalent of hardness producing substance}}$$

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0 Multiplication Factors of Various Salts/Ions:

classmate

Date

Page 5

Dissolved salts/ion. Molar mass chemical equivalent Multiplication factors for converting into equivalents of CaCO_3

1	$\text{Ca}(\text{HCO}_3)_2$	162	81	
2	$\text{Mg}(\text{HCO}_3)_2$	146	73	$100/162$
3	CaSO_4	136	68	$100/146$
4	CaCl_2	111	55.5	$100/136$
5	MgSO_4	120	60	$100/111$
6	MgCl_2	95	47.5	$100/120$
7	CaCO_3	100	50	$100/95$
8	MgCO_3	84	42	$100/100$
9	$\text{Mg}(\text{NO}_3)_2$	148	74	$100/84$
10	Ca^{2+}	40	20	$100/148$
11	Mg^{2+}	24	12	$100/40$
12	HCO_3^-	61	61	$100/24$
13	CO_3^{2-}	60	30	$100/2 \times 61$
14	OH^-	17	17	$100/60$
15	H^+	1	1	$100/2 \times 17$
16	CO_2	44	22	$100/2$
17	HCl	36.5	36.5	$100/44$
18	H_2SO_4	98	49	$100/36.5 \times 2$
19	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	278	139	$100/98$
20 *	$\text{Al}_2(\text{SO}_4)_3$	342	57	$100/278$
21 **	NaAlO_2	82	82	$3 \times 100/342$ or $100/114$ or $50/57$
				$100/82 \times 2$

Q How to calculate CaCO_3 equivalents using multiplication factor

S.No	Substance	Amount of substance(s)	Multiplication factor (M.F)	CaCO_3 equivalent
1.	$\text{Mg}(\text{HCO}_3)_2$	16.8 mg/L	100/146	$\frac{100}{146} \times 16.8 = 11.5$ pp
2.	MgCl_2	19 mg/L	100/95	$\text{M.F} \times \text{S} = 20 \text{ ppm}$
3.	$\text{Mg}(\text{NO}_3)_2$	29.6 ppm	100/148	$\text{M.F} \times \text{S} = 20 \text{ ppm}$
4.	CaCO_3	20 ppm	100/100	$\text{M.F} \times \text{S} = 20 \text{ ppm}$
5.	MgSO_4	24 mg/L	100/120	$\text{M.F} \times \text{S} = 20 \text{ ppm}$
6.	KOH	1.9 ppm	-	-

1. (i)
(ii)
(iii)
(iv)
(v)
(vi)

Solution

Note:

Practice Questions.

1. (c) Magnesium (Mg)

Q. A sample of water on analysis has been found to contain following impurities:-

- (i) Magnesium bicarbonate $\text{Mg}(\text{HCO}_3)_2 \rightarrow 14.6 \text{ mg/l.}$
- (ii) $\text{MgCl}_2 \rightarrow 19 \text{ mg/l.}$
- (iii) $\text{Mg}(\text{NO}_3)_2 \rightarrow 29.6 \text{ mg/l.}$
- (iv) $\text{CaCO}_3 \rightarrow 10 \text{ mg/l.}$
- (v) $\text{MgSO}_4 \rightarrow 36 \text{ mg/l.}$
- (vi) $\text{NaCl} \rightarrow 29 \text{ mg/l.}$

Calculate $\rightarrow$ temporary, permanent & Total hardness of water sample in mg/l & degree clarks.

Solution $\rightarrow$ Temporary hardness is due to the presence of $\text{Mg}(\text{HCO}_3)_2$ & CaCO_3 .

Note:- NaCl , KCl , SiO_2 (silica oxide) there is no hardness due to them, they don't impart hardness.

permanent hardness is due to the MgCl_2 , $\text{Mg}(\text{NO}_3)_2$, $\text{Mg}(\text{NO}_3)_2$ [Magnesium Nitrate], MgSO_4 .

Formula used $\rightarrow \frac{\text{wt of substance} \times 100}{\text{M.W of hardness causing substance}}$

$\rightarrow 1 \text{ ppm} = 1 \text{ mg/l} = 0.07^\circ \text{cl} = 0.1^\circ \text{French.}$

(i) $\text{Mg}(\text{HCO}_3)_2 \Rightarrow \frac{14.6 \times 100}{146} \Rightarrow 10 \text{ mg/l}$

(ii) $\text{CaCO}_3 \rightarrow \frac{10 \times 100}{100} \rightarrow 10 \text{ mg/l}$

$10 + 10 \rightarrow 20 \text{ mg/l}$

$\rightarrow \text{Temporary hardness} \Rightarrow 20 \text{ mg/l} = \boxed{1.40^\circ \text{cl}}$

(iii) $\text{MgCl}_2 \rightarrow \frac{19 \times 100}{95} \Rightarrow 20 \text{ mg/l}$

(iv) $\text{Mg}(\text{NO}_3)_2 \rightarrow \frac{29.6 \times 100}{148} \Rightarrow 20 \text{ mg/l}$

(v) $\text{MgSO}_4 \rightarrow \frac{36 \times 100}{120} \Rightarrow 30 \text{ mg/l}$

permanent hardness $\Rightarrow 70 \text{ mg/l} \Rightarrow \boxed{4.9^\circ \text{cl}}$

$\rightarrow \text{Total hardness} = \text{Temporary} + \text{permanent}$
 $\Rightarrow (1.40 + 4.9)^\circ \text{cl}$
 $\Rightarrow \boxed{6.30^\circ \text{cl}} \text{ Ans.}$

$\rightarrow \text{NaCl}$ doesn't impacts hardness. in given water sample

HW Questions

classmate

Date

Page

5

Q.1 A water sample has following analysis (mg/l)

- (i) $\text{Mg}(\text{HCO}_3)_2 \rightarrow 83 \text{ mg/l}$
- (ii) $\text{Ca}(\text{HCO}_3)_2 \rightarrow 134 \text{ mg/l}$
- (iii) $\text{CaSO}_4 \rightarrow 124 \text{ mg/l}$
- (iv) $\text{MgCl}_2 \rightarrow 84 \text{ mg/l}$
- (v) $\text{CaCl}_2 \rightarrow 94 \text{ mg/l}$
- (vi) $\text{NaCl} \rightarrow 50 \text{ mg/l}$

Calculate temporary, permanent & total hardness in terms of degree clarks.

Q.2 Calculate the total hardness of water in $^\circ\text{F}$ & $^\circ\text{C}$ from the following analysis of water sample.

- $\text{Mg}(\text{HCO}_3)_2 \rightarrow 20.8 \text{ mg/l}$
- $\text{Mg}(\text{NO}_3)_2 \rightarrow 20.8 \text{ mg/l}$
- $\text{MgCl}_2 \rightarrow 21 \text{ mg/l}$
- $\text{MgSO}_4 \rightarrow 20 \text{ mg/l}$
- $\text{CaCO}_3 \rightarrow 21.2 \text{ mg/l}$
- $\text{KCl} \rightarrow 73.2 \text{ mg/l}$

Soln: 1 $\rightarrow$ for temporary

a. $\text{Mg}(\text{HCO}_3)_2 \rightarrow \frac{83 \times 100}{146} = 56.84 \text{ mg/l}$

(b.) $\text{Ca}(\text{HCO}_3)_2 \rightarrow \frac{134 \times 100}{162} = 82.716 \text{ mg/l}$

Temporary $\rightarrow 9.76^\circ\text{C}$

(c) $\text{CaSO}_4 \rightarrow \frac{124 \times 100}{136} \rightarrow 91.17 \text{ mg/l}$

(d) $\text{MgCl}_2 \rightarrow \frac{84 \times 100}{95} \rightarrow 88.42 \text{ mg/l}$

(e) $\text{CaCl}_2 \rightarrow \frac{94 \times 100}{111} \rightarrow 84.684 \text{ mg/l}$

Permanent $\rightarrow 18.49^\circ\text{C}$

2
total $\rightarrow 28.1^\circ\text{C}$ Ans

$\rightarrow$ NaCl doesn't impacts hardness in given water sample.

Solⁿ: 2: $\text{Mg}(\text{HCO}_3)_2 \rightarrow \frac{20.8 \times 100}{162} \rightarrow 12.83 \text{ mg/l.}$

$\text{CaCO}_3 \rightarrow \frac{21.2 \times 100}{100} \rightarrow 21.2 \text{ mg/l.}$

$\text{Mg}(\text{NO}_3)_2 \rightarrow \frac{22.8 \times 100}{148} \rightarrow 22.8 \text{ mg/l}$

$\text{MgCl}_2 \rightarrow \frac{21 \times 100}{95} \rightarrow 22.1 \text{ mg/l.}$

$\text{MgSO}_4 \rightarrow \frac{20 \times 100}{120} \rightarrow 16.6.$

total $\rightarrow 101.53 \text{ mg/l.}$

$\rightarrow 7.710^\circ \text{C}$

$\rightarrow 10.153^\circ \text{F}$ Ans.

21/11/2020

① Determination of hardness of water

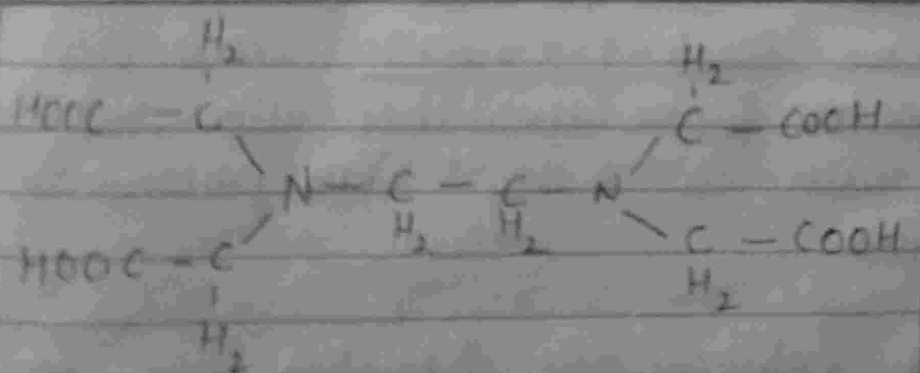
There are two methods which are most commonly used for determination of hardness.

1. Clark's method (soap titration method)
2. EDTA method (Complexometric titration method)

EDTA Method (Complexometric titration method).
(Ethylene diamine tetra acetic Acid).

This method is more accurate for determining the hardness of water mainly caused by dissolved calcium and magnesium salts.

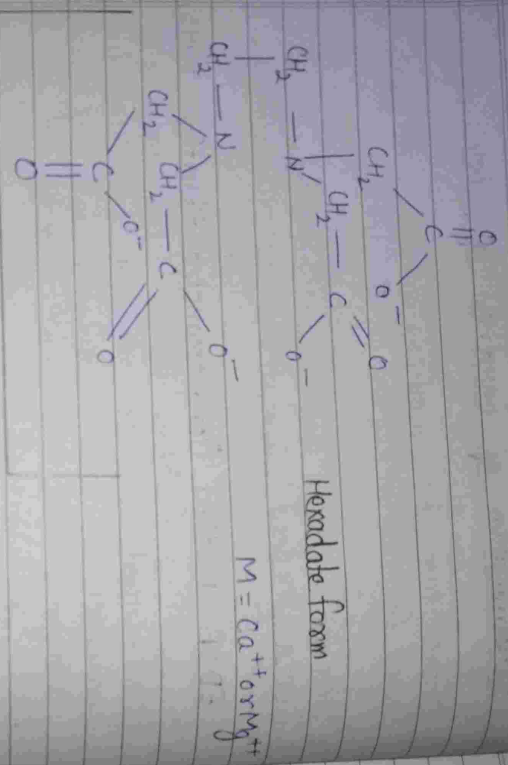
This is a complexometric method since EDTA has limited solubility in water. The disodium salt of EDTA is taken for titration. Ethylene Diamine Tetraacetic Acid (EDTA) is strong complexing agent.



Structure of EDTA.

It acts as a tetra dentate or hexadentate ligand and binds with metal ions to give colourless, soluble and very stable 1:1 complexes (chelates).

Note → Q. Why EDTA is called as Complexometric titration method.
 Ans → Because in this titration EDTA forms complex with the metal present in water sample.



(i) Basic Principle → EDTA forms colorless stable complex with Ca^{2+} and Mg^{2+} ions present in water at pH = 9.8-10.
 → The indicator used in this titration is

Eriochrome Black-T (EBT)
 EBT indicator forms unstable, wine-red colored complex with Ca^{2+} & Mg^{2+} ions present in hard water.

At the end point of the titration, when all the Ca^{2+} & Mg^{2+} ions are consumed, the next drop of EDTA displaces EBT from the weak complex and color changes from wine red to blue due to free EBT in the solⁿ.

Ca^{2+}	EBT Blue
Mg^{2+}	EBT Blue
	(pH = 9-10)

Hard water



Mg

(i) Determine Complex

(ii) Procedure

(A-1) For

(A-2) For

(A-3) For

(A-4) For

(A-1) → Preparation

day

(to use)

solⁿ

Ca²⁺
Mg²⁺ E.B.T Blue
Buffer Soln
(pH=8-10)

[Ca-EBT]
[Mg-EBT]

EDTA

[Ca-EDTA]
[Mg-EDTA]

+ E.B.T (blue)

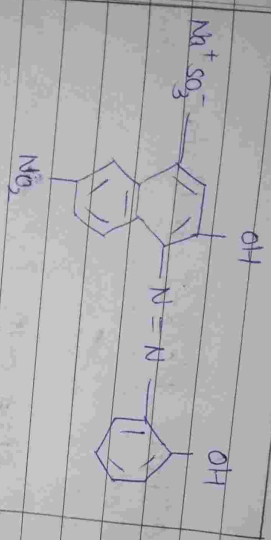
Wine Red colour

Colourless

Hard water

Unstable Complex

Stable Complex



E.B.T
Eriochrome Black-T

i) Determine the hardness of given water sample by complexometric method using EDTA.

(ii) Procedure:- (A) Preparation of solutions:

(A-1) Preparation of standard hard water

(A-2) Preparation of 1/100 EDTA solution

(A-3) Preparation of buffer solution (NH₄Cl-NH₄OH) of pH=10

(A-4) Preparation of Eriochrome Black-T indicator

(A-1) Preparation of Hard water → 10 gram pure & dry CaCl₂ is dissolved in little quantity of H₂O. The soln is then heated till it becomes dry (to remove H₂O & excess acid). Dry residue is then dissolved in distilled water & to make one litre soln 1ml standard hard water = 1mg CaCl₂.

(A-2) Preparation of $M/100$ EDTA soln: $\rightarrow$ 3.722 g of disodium salt of EDTA is dissolved in distilled water to make up to 1 litre soln.

(A-3) Preparation of $\text{NaCl-NH}_4\text{OH}$ of $\text{pH}-10$: $\rightarrow$ 61.5 gm of A.R. NH_4Cl & 570 ml concentrated ammonia soln is taken and diluted to 1 litre with de-ionized water.

(A-4) Preparation of Eriochrome Black-T indicator: $\rightarrow$ 0.5 gm of Eriochrome Black-T indicator is dissolved in 100 ml ethyl alcohol.

(B) Standardization of EDTA soln. (Water sample - Hard water)
50 ml of standard hard water is taken into a conical flask, 8-10 ml of buffer solution and 3-4 drops of In^{EBT} is added to it. then it is titrated with EDTA till wine red colour of the mixture changes to blue. This is end point. Let the volume of EDTA soln consumed here is V_1 ml.

(C) Determination of total hardness: $\rightarrow$ 50 ml of ~~std sample hard~~ standard tap water is taken in flask, 8-10 ml of buffer soln & 3-4 drops of indicator EBT is added. then it is titrated till wine red color of mix. changes to Blue. this is end point. let the vol^m of EDTA soln consumed is V_2 ml.

(D) Determination sample hard. The precipitate with distilled water washings and adding water titrated vol^m of

(iii) calculation

g of
distilled

5 gm of
soda
with

0.5 gm
in

and water
to

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is

of

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(D) Determination of permanent hardness: $\rightarrow$ 250 ml of sample hard water is boiled till it reduces to 50 ml. The precipitate is filtered off and is washed with distilled water. Both the filtrate and washings are collected in a 250 ml capacity flask and volume is again made 250 ml by adding distilled water. 50 ml of this boiled water is taken into a conical flask and it is titrated with EDTA soln same as above. Let the volm of EDTA soln consumed here is V_3 ml.

(iii) Calculations:—